MODULE 4 RESPIRATORY SYSTEM

The Respiratory System.

THE RESPIRATORY SYSTEM

A reference for the respiratory system video and lab. This chapter covers the upper and lower respiratory tracts, the larynx, the bronchial tree, the lungs and pleurae, the alveoli, and common respiratory disorders. The focus is on structure.

BY THE END

1. Distinguish the upper and lower respiratory tracts and the conducting and respiratory zones.
2. Identify the parts of the nasal cavity and the pharynx.
3. Identify the cartilages of the larynx and the parts of the bronchial tree.
4. Describe the lungs, the pleurae, and the structure of an alveolus.
5. Name common respiratory disorders.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The respiratory system

drsrennie-stack.github.io/new-build-bio4-solano/respiratory-concept-videos.html

The Respiratory System, an Overview

The respiratory system moves air in and out and brings it close to the blood. Its parts are grouped two ways, and the two groupings do not line up, which is exactly why both are tested.

THE TWO WAYS OF GROUPING THE PARTS

Structure	What it is
Respiratory system	The organs that move air and bring it close to the blood for gas exchange
Upper respiratory tract	The nose, nasal cavity, paranasal sinuses, and pharynx
Lower respiratory tract	The larynx, trachea, bronchi, and lungs
Conducting zone	The passages that only carry air, from the nose to the terminal bronchioles. It warms, moistens, and filters
Respiratory zone	The structures where gas exchange happens: the respiratory bronchioles, alveolar ducts, and alveoli
A note on scope	This chapter covers the structure of the system. The exchange of gases is a physiology topic and is not covered here

The Upper Respiratory Tract

The upper tract takes in air and conditions it, warming, moistening, and filtering it before it reaches the lungs.

THE NOSE AND NASAL CAVITY

Structure	What it is
External nose	The visible part, framed by the nasal bones and the frontal process of the maxilla above and hyaline cartilage below
Root	The area between the eyebrows, where the nose meets the forehead
Bridge	The upper bony part of the nose
Dorsum nasi	The anterior margin, running from root to apex
Apex	The tip of the nose
Nares	The external openings, the nostrils
Nasal cavity	The space behind the nose, divided into right and left halves by the nasal septum
Nasal septum	Septal cartilage in front, the vomer and the perpendicular plate of the ethmoid behind
Nasal vestibule	The area just inside the nostrils, lined with skin, sebaceous glands, and coarse hairs that filter particles
Nasal conchae	The three scroll-shaped projections on the lateral wall that swirl the air and increase contact with the warm, moist lining
Nasal meatuses	The passages beneath each concha, named superior, middle, and inferior
Hard palate	The bony floor of the nasal cavity, separating it from the oral cavity
Soft palate	The muscular posterior floor, which lifts to close the nasopharynx during swallowing
Choanae	The posterior openings where the nasal cavity becomes the nasopharynx
Paranasal sinuses	Air-filled cavities in the frontal, ethmoid, sphenoid, and maxillary bones that drain into the nasal cavity

The mucosa of the nasal cavity

TWO MUCOSAE, TWO JOBS

Mucosa	Epithelium	Role
Olfactory mucosa	Olfactory epithelium, in a small patch in the roof	Houses the receptors for smell
Respiratory mucosa	Pseudostratified ciliated columnar with goblet cells	Secretes mucus, traps particles, and sweeps them toward the pharynx to be swallowed

HOLD ONTO THIS

Cold air makes the cilia sluggish, so the mucus stops moving backward and runs forward instead. That is why a nose runs in cold weather, and it is a structural answer to an everyday question.

The pharynx

THE THREE REGIONS, TOP TO BOTTOM

Region	Location	Carries	Epithelium and structures
Nasopharynx	Behind the nasal cavity, above the soft palate	Air only	Pseudostratified ciliated columnar. Holds the pharyngeal tonsil and the openings of the auditory tubes
Oropharynx	Behind the oral cavity, from the soft palate to the epiglottis	Air and food	Stratified squamous, for friction. Holds the palatine and lingual tonsils
Laryngopharynx	From the epiglottis to the esophagus	Air and food	Stratified squamous. The point where the air and food paths divide

The Larynx

The larynx, the voice box, connects the pharynx to the trachea. It keeps food out of the airway, holds the airway open, and houses the vocal cords. Nine cartilages build it, three single and three paired.

THE NINE CARTILAGES

Cartilage	Number	Description
Thyroid cartilage	Single	The large shield-shaped cartilage forming the laryngeal prominence, the Adam's apple
Cricoid cartilage	Single	The ring-shaped cartilage just below the thyroid cartilage, the only complete ring in the airway
Epiglottis	Single	The flexible spoon-shaped flap of elastic cartilage that folds over the laryngeal inlet during swallowing
Arytenoid cartilages	Paired	Anchor the vocal folds and move them. The most important of the paired cartilages
Corniculate cartilages	Paired	Sit on top of the arytenoids
Cuneiform cartilages	Paired	Lie in the fold between the arytenoid and the epiglottis

THE FOLDS AND THE OPENING

Structure	What it is
Vocal folds	The true vocal cords. Pearly white and avascular, they vibrate as air passes to produce sound
Vocal ligaments	The elastic bands within the vocal folds, running from the arytenoid cartilages to the thyroid cartilage
Vestibular folds	The false vocal cords, above the true folds. They close the airway during swallowing and play no part in sound
Glottis	The vocal folds together with the opening between them

HOLD ONTO THIS

Every laryngeal cartilage is hyaline except the epiglottis, which is elastic. It has to bend and spring back with every swallow, and hyaline cartilage would not.

The Trachea and Bronchial Tree

Below the larynx the airway becomes a branching tree, growing narrower and more numerous at every division. Follow the path of air in order.

1. **Trachea** the windpipe, held permanently open by C-shaped rings of hyaline cartilage, with the trachealis muscle closing the gap posteriorly against the esophagus
2. **Carina** the internal ridge where the trachea forks into the two main bronchi
3. **Main bronchi** the primary bronchi, one entering each lung. The right is wider and more vertical
4. **Lobar bronchi** the secondary bronchi, one to each lobe of the lung
5. **Segmental bronchi** the tertiary bronchi, one to each bronchopulmonary segment
6. **Bronchioles** the smallest conducting airways, with smooth muscle walls and no cartilage
7. **Terminal bronchioles** the last airways of the conducting zone, just before gas exchange begins

HOLD ONTO THIS

Cartilage decreases and smooth muscle increases as the tree gets smaller. The switch point is the bronchiole, which is why bronchioles can constrict and bronchi cannot, and why asthma is a disease of the small airways.

The Lungs and Pleurae

The two lungs fill most of the thoracic cavity. They are not identical: the left lung gives up room for the heart.

COMPARE THE TWO LUNGS

Lung	Lobes	Fissures	Note
Right lung	Three: superior, middle, and inferior	A horizontal fissure and an oblique fissure	The larger of the two
Left lung	Two: superior and inferior	An oblique fissure only	Smaller, with a cardiac notch that makes room for the heart

THE SURFACES AND THE COVERING

Structure	What it is
Apex	The narrow superior tip of a lung, rising just above the clavicle
Base	The broad inferior surface of a lung, resting on the diaphragm
Hilum	The indentation on the medial surface where the bronchus and vessels enter and leave
Root of the lung	The bundle at the hilum: main bronchus, pulmonary artery, pulmonary veins, bronchial vessels, nerves, and lymphatics
Parietal pleura	The serous membrane lining the thoracic wall
Visceral pleura	The serous membrane covering the outer surface of the lung
Pleural cavity	The thin, fluid-filled space between the two pleurae that lets the lung glide as it expands
Cardiac notch	The indentation in the left lung that accommodates the heart
Lingula	The tongue-shaped projection of the left superior lobe, below the cardiac notch

The Respiratory Zone and the Alveoli

Past the terminal bronchioles the airway becomes the respiratory zone, where the walls thin out into air sacs and gas exchange becomes possible.

THE STRUCTURES OF THE RESPIRATORY ZONE

Structure	What it is
Respiratory bronchioles	The first airways of the respiratory zone, with a few alveoli budding from their walls
Alveolar ducts	Passages whose walls are lined entirely with alveoli
Alveolar sac	A cluster of alveoli that share a common space
Alveoli	The tiny air sacs that are the site of gas exchange. The lungs hold hundreds of millions of them
Type I alveolar cells	The thin, flat simple squamous cells that form most of the alveolar wall, where gases cross
Type II alveolar cells	The cuboidal cells that secrete surfactant, which lowers surface tension and keeps the alveoli from collapsing
Alveolar macrophages	The dust cells that patrol the alveolar surface and engulf debris
Respiratory membrane	The thin barrier between the alveolar air and the blood: alveolar epithelium, fused basement membranes, and capillary endothelium
Pulmonary capillaries	The dense net of capillaries wrapping each alveolus

Respiratory Disorders

Respiratory disorders mostly block airflow or damage the surface where gases cross.

COMPARE THE COMMON ONES BY THE STRUCTURE EACH ONE AFFECTS

Disorder	Structure affected	What it is
Asthma	The bronchioles	Inflammation and narrowing of the bronchioles, causing wheezing and breathlessness
COPD	The airways and alveoli	Chronic obstructive pulmonary disease, a long-term blockage of airflow that includes emphysema and chronic bronchitis
Emphysema	The alveoli	Destruction of the alveolar walls, which reduces the surface available for gas exchange
Pneumonia	The alveoli	An infection that inflames the alveoli and fills them with fluid
Pneumothorax	The pleural cavity	Air in the pleural cavity, which breaks the seal and lets the lung collapse
Lung cancer	The lung tissue	A malignant tumor of the lung, strongly linked to smoking

WHY THE DESIGN HAS WEAK POINTS

The **epiglottis** normally seals the airway during a swallow. When something slips past it, a coughing fit is the airway's attempt to clear it. Because the right main bronchus is wider and more vertical than the left, an inhaled object most often lands in the right lung.

The **alveoli** give the lungs an enormous surface, but it is fragile. Emphysema destroys alveolar walls and merges many small sacs into a few large ones, trading a huge exchange surface for a small one. The change is structural and permanent.

The **pleural cavity** is sealed and holds only a thin film of fluid. If the chest wall or lung is punctured, air enters that space, the seal is lost, and the lung collapses, a pneumothorax. The structure that lets the lung glide is also what makes it vulnerable.

The **cricoid cartilage** is the only complete ring in the airway, which is why the space just above it, the cricothyroid membrane, is the landmark for an emergency airway. Anatomy chooses that spot, not convenience.